

Number System Conversions

① Convert the following decimal number to binary

a) 18

$$\begin{array}{r}
 2 \overline{) 18} \quad 0 \\
 2 \overline{) 9} \quad 1 \\
 2 \overline{) 4} \quad 0 \\
 2 \overline{) 2} \quad 0 \\
 \hline
 1
 \end{array}$$

$$18_{(10)} = 10010_{(2)}$$

b) 45

$$\begin{array}{r}
 2 \overline{) 45} \quad 1 \\
 2 \overline{) 22} \quad 0 \\
 2 \overline{) 11} \quad 1 \\
 2 \overline{) 5} \quad 1 \\
 2 \overline{) 2} \quad 0 \\
 \hline
 1
 \end{array}$$

$$45_{(10)} = 101101_{(2)}$$

② Convert the following binary number to decimal

a) 10101

$$\begin{array}{cccccc}
 1 & 0 & 1 & 0 & 1 & \\
 16 & 8 & 4 & 2 & 1 & \\
 16 + 4 + 1 & = & 21 & & &
 \end{array}$$

$$(10101)_{(2)} = 21_{(10)}$$

b) 110011

$$\begin{array}{cccccc}
 1 & 1 & 0 & 0 & 1 & 1 \\
 32 & 16 & 8 & 4 & 2 & 1 \\
 32 + 16 + 2 + 1 & = & 51 & & &
 \end{array}$$

$$(110011)_{(2)} = 51_{(10)}$$

③ Convert the following decimal numbers to octal

a) 74

$$\begin{array}{r} 8 \overline{) 74} \quad 2 \\ 8 \overline{) 9} \quad 1 \\ 1 \end{array}$$

$$74_{(10)} = 112_{(8)}$$

b) 255

$$\begin{array}{r} 8 \overline{) 255} \quad 7 \\ 8 \overline{) 31} \quad 7 \\ 3 \end{array}$$

$$255_{(10)} = 377_{(8)}$$

④ Convert the following octal numbers to decimal

a) $(127)_8$

$$\begin{array}{ccc} 1 & 2 & 7 \\ 64 & 2 \times 8 & 7 \times 1 \\ 64 & 16 & 7 \end{array}$$

$$64 + 16 + 7 = 87$$

$$(127)_8 = 87_{(10)}$$

b) $(345)_8$

$$\begin{array}{ccc} 3 & 4 & 5 \\ 3 \times 64 & 4 \times 8 & 5 \times 1 \\ 192 & 32 & 5 \end{array}$$

$$192 + 32 + 5 = 229$$

$$345_{(8)} = 229_{(10)}$$

⑤ Convert the following hexadecimal number to decimal

a) $(1A)_{16}$

$$\begin{array}{cc} 1 & A \\ 1 & 10 \\ 1 \times 16 & 10 \times 1 \\ 16 & 10 \end{array}$$

$$16 + 10 = 26$$

$$(1A)_{16} = 26_{(10)}$$

b) $(2F)_{16}$

$$\begin{array}{cc} 2 & F \\ 2 & 15 \\ 2 \times 16 & 15 \times 1 \\ 32 & 15 \end{array}$$

$$32 + 15 = 47$$

$$(2F)_{16} = 47_{(10)}$$

⑥ Convert $(255)_{10}$ to binary, octal, hexadecimal

$$(265)_{10}$$

Binary

Octal

Hexadecimel

$$\begin{array}{r} 2 \overline{) 265} \quad 1 \\ 2 \overline{) 127} \quad 1 \\ 2 \overline{) 63} \quad 1 \\ 2 \overline{) 31} \quad 1 \\ 2 \overline{) 15} \quad 1 \\ 2 \overline{) 7} \quad 1 \\ 2 \overline{) 3} \quad 1 \\ \hline \end{array}$$

$$\begin{array}{r} 8 \overline{) 25537} \\ \underline{8 317} \\ 3 \end{array}$$

$$16 \overline{) 255} \quad 15$$

$$(FP)_{16}$$

$$(377)_{(8)}$$

$$(11111111)_2$$

⑦ Convert $(1101011)_2$ to octal & hexa decimal

$$(1101011)_2$$

~~$$\begin{array}{ccccccc}
 1 & 1 & 0 & 1 & 0 & 1 & 1 \\
 301 & 16 & 3 & 4 & 2 & 1 & 0 \\
 32 & 16 & 4 & 1 & 2 & 5 & 3
 \end{array}$$~~

11 0 1 0 1 1
64 32 16 8 4 2 1

$$64 + 32 + 8 + 241$$

$$= 107$$

$$(1101011)_2 = (107)_{10}$$

Octal

$$\begin{array}{r|l} 8 & 107 \quad 3 \\ \hline 8 & 13 \quad 5 \\ \hline & 1 \end{array}$$

$$153_{(8)}$$

Hexadecimal

$$\begin{array}{r|l} 16 & 107 \quad 11 \\ \hline & 6 \end{array}$$

$$6B_{(16)}$$

⑧ Convert $(3A5)_{16}$ to binary and decimal

$$(3A5)_{16}$$

Decimal

$$3 \quad A \quad 5$$

$$3 \quad 10 \quad 5$$

$$3 \times 256 \quad 10 \times 16 \quad 5 \times 1$$

$$768 + 160 + 5 = 933$$

$$933_{(10)}$$

Binary

~~$$\begin{array}{r|l} 2 & 933 \\ \hline 2 & 466 \quad 0 \\ \hline 2 & 233 \quad 1 \\ \hline 2 & 116 \quad 0 \\ \hline 2 & 58 \quad 0 \\ \hline 2 & 29 \quad 1 \\ \hline 2 & 14 \quad 0 \\ \hline 2 & 7 \quad 1 \\ \hline 2 & 3 \quad 1 \\ \hline & 1 \end{array}$$~~

$$\begin{array}{r|l} 2 & 933 \quad 1 \\ \hline 2 & 466 \quad 0 \\ \hline 2 & 233 \quad 1 \\ \hline 2 & 116 \quad 0 \\ \hline 2 & 58 \quad 0 \\ \hline 2 & 29 \quad 1 \\ \hline 2 & 14 \quad 0 \\ \hline 2 & 7 \quad 1 \\ \hline 2 & 3 \quad 1 \\ \hline & 1 \end{array}$$

$$(1110100101)_2$$